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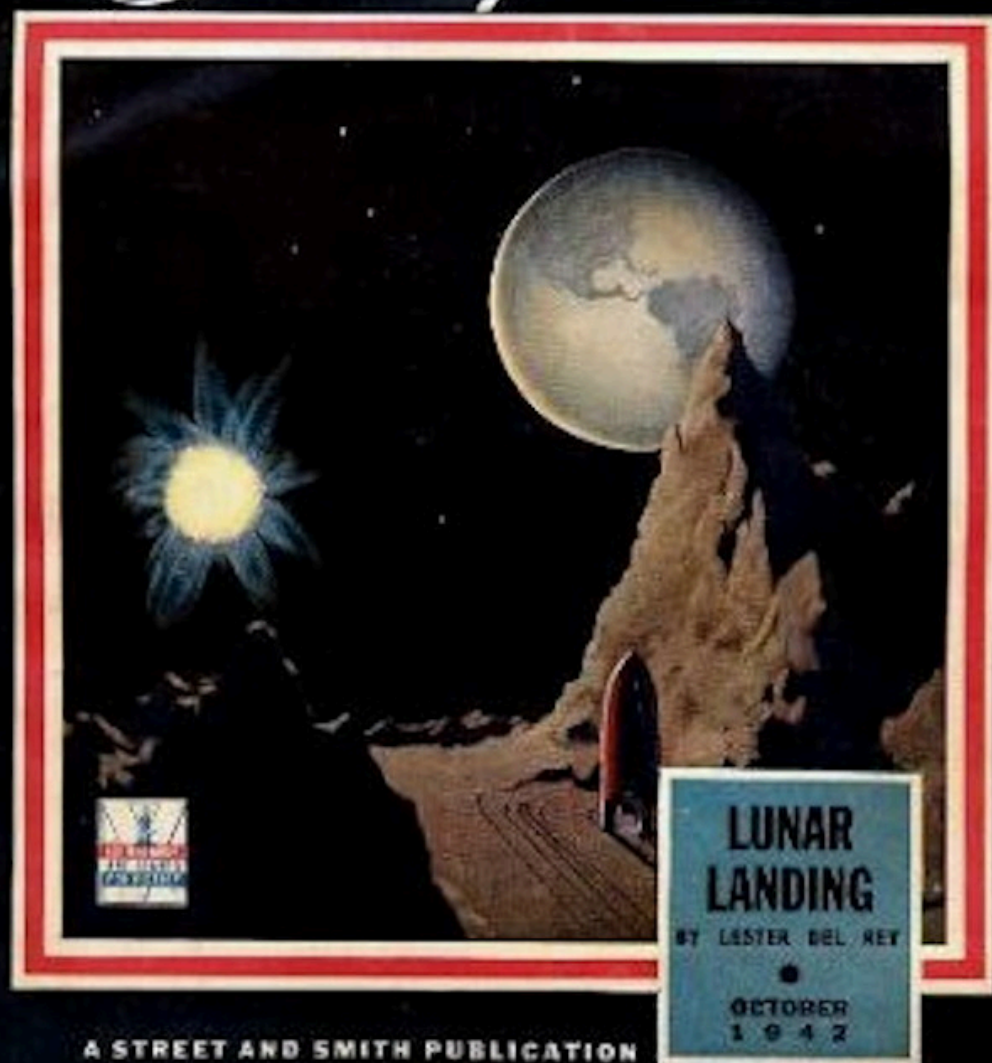
LUNAR LANDING

BY LESTER DEL REY

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*** START OF THE PROJECT GUTENBERG EBOOK QRM-
INTERPLANETARY ***

QRM—INTERPLANETARY

By George O. Smith

Illustrated by Kolliker

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QRM—International code signal meaning "Interference" of controllable nature, such as man-made static, cross modulation from another channel adjoining or willful obliteration of signals by an interfering source.

Interference not of natural sources (designated by International code as QRN), such as electrical storms, common static, et cetera.

—Handbook, Interplanetary Amateur Radio League.

Korvus, the Magnificent, Nilamo of Yoralen, picked up the telephone in his palace and said: "I want to talk to Wilneda. He is at the International Hotel in Detroit, Michigan."

"I'm sorry, sir," came the voice of the operator. "Talking is not possible, due to the fifteen-minute transmission lag between here and Terra. Interplanetary Communications will not permit audio. However, teletype messages are welcome."

Her voice originated fifteen hundred miles north of Yoralen, but it sounded as though she might be in the next room. Korvus thought for a moment and then said: "Take this message: 'Wilneda: Add to order for mining machinery one type 56-XXD flier to replace washed-out model. And remember, tobacco and sublevel energy will not mix!' Sign that *Korvus*."

"Yes, Mr. Korvus."

"Not *mister*!" yelled the monarch. "I am Korvus, the Magnificent! I am Nilamo of Yoralen!"

"Yes, your magnificence," said the operator humbly. It was more than possible that she was stifling a laugh, which knowledge made the little man of Venus squirm in wrath. But there was nothing that he could do about it, television still being distant by the same five years that it was behind in 1929.

To give Korvus credit, he was not a pompous little man. He was large—for a Venusian—which made him small according to the standards set up by Terrestrials. He, as Nilamo of Yoralen, had extended the once-small

kingdom outward to include most of the Palanortis Country which extended from about 23.0 degrees North Latitude to 61.7 degrees, and almost across the whole, single continent that was the dry land of Venus. He was a wily monarch, making his conquest of the wild and lawless country by treaty, and by double-double-crossing those who might have tried to double-cross him. Armed conquest was scorned, but armed defense was desirable in the Palanortis Country—and Korvus had defended himself up and down the inhabitable Northern portion of the planet. His conquest had been a blessing to civilization, and though publicly denounced, it was privately commended. Those who could have stopped it did all they could to delay and intercept any proceedings that would have caused the conquest of Korvus' intended country any trouble.

Korvus' message to Terra zoomed across the fifteen hundred rocky miles of Palanortis to Northern Landing. It passed high across the thousand-foot-high trees and over the mountain ranges. It swept over open patches of water, and across intervening cities and towns. It went with the speed of light and in a tight beam from Yoralen to Northern Landing, straight as a die and with person-to-person clarity. The operator in the city that lay across the North Pole of Venus clicked on a teletype, reading back the message as it was written.

Korvus told him: "That is correct."

"The message will be in the hands of your representative Wilneda within the hour!"

The punched tape from Operator No. 7's machine slid along the line. It entered a coupling machine and was stripped from the tape and repunched upon a tape that was operating at better than a thousand words per minute. Operator 7's tape then left the machine to be rolled into a file roll and placed in the vaults below the city. It was of no use save as a reference from now on.

The coupling machine worked furiously. It accepted the tapes from seventy operators as fast as they could write them. It selected the messages as they entered the machine, placing a mechanical preference upon whichever

message happened to be ahead of the others on the moving tapes. The master tape moved continuously at eleven hundred words per minute, taking teletype messages from everywhere in the Northern Hemisphere of Venus to any of the other planets in the Solar System. It was a busy machine, for even at eleven hundred words per minute, it often got hours behind.

The synchronous-keyed signal from the coupling machine left the operating room and went to the transmission room. It was amplified and hurled out of the city to a small, squat building at the outskirts of Northern Landing.

Here it was impressed upon a carrier wave and flung at the sky.

But not alone. Not unguarded. The upper half of the building carried a monstrous parabolic reflector, mounted on gimbals. The signal was focused into a beam. The beam was made of two components. The center component was a circularly polarized, ultra-high frequency wave of five centimeter waves, modulated with the keying signals of the teletype coupling machine. The outer component was a radially polarized wave of one centimeter waves. A radio frequency armor.

It was hurled at the sky, a concentric wave, out of a reflector, by a thousand kilowatt transmitter. The wave seared against the Venusian Heaviside Layer. The outer component bored at the ionization. It chewed and it bit. It fought and it struggled. It destroyed ionization by electronically shorting the ionization. And, as is the case with strife, it lost heavily in the encounter. The beam was resisted fiercely. Infiltrations of ionization tore at the central component, stripping and trying to beat it down.

But man triumphed over nature! The megawatt of energy that came in a tight beam from the building at Northern Landing emerged from the Heaviside Layer as a weak, piffling signal. It wavered and it crackled. It wanted desperately to lie down and sleep. Its directional qualities were impaired, and it wobbled badly. It arrived at the relay station tired and worn.

One million watts of ultra-high frequency energy at the start, it was measurable in microvolts when it reached the relay station, only five hundred miles above the city of Northern Landing.

The signal, as weak and as wobbly as it was, was taken in by eager receptors. It was amplified. It was dehashed, de-staticked, and deloused.

And once again, one hundred decibels stronger and infinitely cleaner, the signal was hurled out on a tight beam from a gigantic parabolic reflector.

Across sixty-seven million miles of space went the signal. Across the orbit of Venus it went in a vast chord. It arrived at the Venus Equilateral Station with less trouble than the original transmission through the Heaviside Layer. The signal was amplified and demodulated. It went into a decoupler machine where the messages were sorted mechanically and sent, each to the proper channel, into other coupler machines. Beams from Venus Equilateral were directed at Mars and at Terra.



The Terra beam ended at Luna. Here it again was placed in the two-component beam and from Luna it punched down at Terra's Layer. It emerged into the atmosphere of Terra, as weak and as tired as it had been when it had come out of the Venus Layer. It entered a station in the Bahamas, was stripped of the interference, and put upon the land beams. It entered decoupling machines that sorted the messages as to destination. These various beams spread out across the face of Terra, the one carrying Korvus' message finally coming into a station at Ten Mile Road and Woodward. From this station at the outskirts of Detroit, it went upon land wires downtown to the International Hotel.

The teletype machine in the office of the hotel began to click rapidly. The message to Wilneda was arriving.

And fifty-five minutes after the operator told Korvus that less than an hour would ensue, Wilneda was saying, humorously, "So, Korvus was drunk again last night—"

Completion of Korvus' message to Wilneda completes also one phase of the tale at hand. It is not important. There were a hundred and fifty other messages that might have been accompanied in the same manner, each as interesting to the person who likes the explanation of the interplanetary communication service. But this is not a technical journal. A more complete explanation of the various phases that a message goes through in leaving a city on Venus to go to Terra may be found in the Communications Technical Review, Volume XXVII, Number 8, pages 411 to 716. Readers more interested in the technical aspects are referred to the article.

But it so happens that Korvus' message was picked out of a hundred-odd messages because of one thing only. At the time that Korvus' message was in transit through the decoupler machines at Venus Equilateral Relay Station, something of a material nature was entering the air lock of the station.

It was an unexpected visit.

Don Channing looked up at the indicator panel in his office and frowned in puzzlement. He punched a buzzer and spoke into the communicator on his desk.

"Find out who that is, will you, Arden?"

"He isn't expected," came back the voice of Arden Westland.

"I know that. But I've been expecting someone ever since John Walters retired last week. You know why."

"You hope to get his job," said the girl in an amused voice. "I hope you do. So that someone else will sit around all day trying to make you retire so that he can have your job!"

"Now look, Arden, I've never tried to make Walters retire."

"No, but when the word came that he was thinking of it, you began to think about taking over. Don't worry, I don't blame you." There was quite a protracted silence, and then her voice returned. "The visitor is a gentleman by the name of Francis Burbank. He came out in a flitter with a chauffeur and all."

"Big shot, hey?"

"Take it easy. He's coming up the office now."

"I gather that he desires audience with me?" asked Don.

"I think that he is here to lay down the law! You'll have to get out of Walters' office, if his appearance is any guide."

There was some more silence. The communicator was turned off at the other end, which made Channing fume. He would have preferred to hear the interchange of words between his secretary and the newcomer. Then, instead of having the man announced, the door opened abruptly and the stranger entered. He came to the point immediately.

"You're Don Channing? Acting Director of Interplanetary Communications?"

"I am."

"Then I have some news for you, Dr. Channing. I have been appointed Director by the Interplanetary Communications Commission. You are to resume your position as Electronics Engineer."

"Oh?" said Channing. His face fell. "I sort of believed that I would be offered that position."

"There was a discussion of that procedure. However, the Commission decided that a man of more commercial training would better fill the position. The Communications Division has been operating at too small a profit. They felt that a man of commercial experience could cut expenses and so on to good effect. You understand their reasoning, of course," said Burbank.

"Not exactly."

"Well, it is like this. They know that a scientist is not usually a man to consider the cost of experimentation. They build thousand-ton cyclotrons to convert a penny's worth of lead into one and one-tenth cents' worth of lead and gold. And they use three hundred dollars' worth of power and a million-dollar machine to do it with.

"They feel that a man with training like that will not know the real meaning of the phrase, 'cutting expenses.' A new broom sweeps clean, Dr. Channing. There must be many places where a man of commercial experience can cut expenses. I, as Director, shall do so."

"I wish you luck," said Channing.

"Then there is no hard feeling?"

"I can't say that. It is probably not your fault. I cannot feel against you, but I do feel sort of let down at the decision of the Commission. I have had experience in this job."

"The Commission may appoint you to follow me. If your work shows a grasp of commercial operations, I shall so recommend."

"Thanks," said Channing dryly. "May I buy you a drink?"

"I never drink. And I do not believe in it. If it were mine to say, I'd prohibit liquor from the premises. Venus Equilateral would be better off without it."

Don Channing snapped the communicator. "Miss Westland, will you come in?"

She entered, puzzlement on her face.

"This is Mr. Burbank. His position places him in control of this office. You will, in the future, report to him directly. The report on the operations, engineering projects, and so on that I was to send in to the Commission this morning will, therefore, be placed in Mr. Burbank's hands as soon as possible."

"Yes, Dr. Channing." Her eyes held a twinkle, but there was concern and sympathy in them, too. "Shall I get them immediately?"

"They are ready?"

"I was about to put them on the tape when you called."

"Then give them to Mr. Burbank." Channing turned to Burbank. "Miss Westland will hand you the reports I mentioned. They are complete and precise. A perusal of them will put you in grasp of the situation here at Venus Equilateral better than will an all-afternoon conference. I'll have Miss Westland haul my junk out of here. You may consider this as your office, it having been used by Walters. And, in the meantime, I've got to check up on some experiments down on the ninth level." Channing paused, "You'll excuse me?"

"Yes, if Miss Westland knows where to find you."

"She will. I'll inform her of my whereabouts."

"I may want to consult you after I read the reports."

"That will be all right. The autocall can find me anywhere on Venus Equilateral, if I'm not at the place Miss Westland calls."

Don Channing stopped at Arden's desk. "I'm booted," he told her.

"Leaving Venus Equilateral?" she asked with concern.

"No, blond and beautiful, I'm just shunted back to my own office."

"Can't I go with you?" pleaded the girl.

"Nope. You are to stay here and be a nice, good-looking Mata Hari. This bird seems to think that he can run Interplanetary Communications like a bus or a factory. I know the type, and the first thing he'll do is to run Interplanetary Communications into a snarl. Keep me informed of anything complicated, will you?"

"Sure. And where are you going now?"

"I'm going down and get Walt Franks. We're going to inspect the transparency of a new type of glass."

"I didn't know that optical investigations come under your jurisdiction."

"This investigation will consist of a visit to the ninth level."

"Can't you take me along?"

"Not today," he grinned. "Your new boss does not believe in the evils of looking through the bottom of a glass. We must behave with decor. We must forget fun. We are now operating under a man who will commercialize electronics to a fine art."

"Don't get stewed. He may want to know where the electrons are kept."

"I'm not going to drink that much. Walt and I need a discussion," he said. "And in the meantime, haul my spinach out of the office, will you, and take it back to the electronics office. I'll be needing it back there."

"O.K., Don," she said. "I'll see you later."

Channing left to go to the ninth level. He stopped long enough to collect Walt Franks.

Over a tall glass of beer, Channing told Franks of Burbank's visit. And why.

Only one thing stuck in Franks' mind. "Did you say that he might close Joe's?" asked Franks.

"He said that if it were in his power to do so, he would."

"Heaven forbid. Where will we go to be alone?"

"Alone?" snorted Channing. The barroom was half filled with people, being the only drinking establishment for sixty-odd million miles.

"Well, you know what I mean."

"I could smuggle in a few cases of beer," suggested Don.

"Couldn't we smuggle him out?"

"That would be desirable. But I think he is here to stay. Darn it all, why do they have to appoint some confounded political pal to a job like this? I'm telling you, Walt, he must weigh two hundred if he weighs a pound. He holds his stomach on his lap when he sits down."

Walt looked up and down Channing's slender figure. "Well, he won't be holding Westland on his lap if it is filled with stomach."

"I never hold Westland on my lap—"

"No?"

"—during working hours!" finished Channing. He grinned at Franks and ordered another beer. "And how is the Office of Beam Control going to make out under the new regime?"

"I'll answer that after I see how the new regime treats the Office of Beam Control," answered Franks. "I doubt that he can do much to bugger things up in my office. There aren't many cheaper ways to direct a beam, you know."

"Yeah. You're safe."

"But what I can't understand is why they didn't continue you in that job. You've been handling the business ever since last December when Walters got sick. You've been doing all right."

"Doing all right just means that I've been carrying over Walters' methods and ideas. What the Commission wants, apparently, is something new. Ergo the new broom."

"Personally, I like that one about the old shoes being more comfortable," said Franks. "If you say the word, Don, I'll slip him a dose of high voltage. That should fix him."

"I think that the better way would be to work for the bird. Then when he goes, I'll have his recommendation."

"Phooey," snorted Franks. "They'll just appoint another political pal. They've tried it before and they'll try it again. I wonder what precinct he carries."

The telephone rang in the bar, and the bartender, after answering, motioned to Walt Franks. "You're wanted in your office," said the bartender. "And besides," he told Channing, "if I'm going to get lunch for three thousand people, you'd better trot along, too. It's nearly eleven o'clock, you know, and the first batch of five hundred will be coming in."

He wasn't quite accurate as to the figures. The complement of Venus Equilateral was just shy of twenty-seven hundred. They worked in three eight-hour shifts, about nine hundred to a shift. They had their lunch and dinner hours staggered so that at no time was there more than about two hundred people in the big lunchroom. The bar, it may be mentioned, was in a smaller room at one end of the much larger cafeteria.

Venus Equilateral Relay Station was a modern miracle of engineering if you liked to believe the books. Actually, Venus Equilateral was an asteroid that had been shoved into its orbit about the Sun, forming a practical demonstration of the equilateral triangle solution of the Three Moving Bodies. It was a long cylinder, about three miles in length by about a mile in diameter.

There was little of the original asteroid. At the present time, most of the original rock had been discarded to make room for the ever-growing personnel and material that were needed to operate the relay station. What had been an asteroid with machinery was now a huge pile of machinery with people. The insides, formerly of spongy rock, were now neatly cubed off into offices, rooms, hallways, and so on, divided by sheets of steel. The outer surface, once rugged and forbidding, was now almost all shiny steel. The small asteroid, a tiny thing, was far smaller than the present relay station, the station having overflowed the asteroid soon after men found that uninterrupted communication was possible between the worlds.

Now, the man-made asteroid carried twenty-seven hundred people. There were stores, offices, places of recreation, churches, marriages, deaths, and

everything but taxes. Judging by its population, it was a small town.

Venus Equilateral rotated about its axis. On the inner surface of the shell were the homes of the people—not cottages, but apartmental cubicles, one, two, three, six rooms. The rotation made a little more than one Earth G of artificial gravity. Above this outer shell of apartments, the offices began. Offices, recreation centers, and so on. Up in the central portion where the gravity was nil or near-nil, the automatic machinery was placed. The gyroscopes and the beam finders, the storerooms, the air plants, the hydroponic farms, and all other things that needed little or no gravity for well-being.

This was the Venus Equilateral Relay Station, sixty degrees ahead of the planet Venus, on Venus' orbit. Often closer to Terra than Venus, the relay station offered a perfect place to relay messages through whenever Mars or Terra were on the other side of the Sun. It was seldom idle, for it was seldom that both Mars and Terra were in such position that direct communication between the three planets was possible.

This was the center of Interplanetary Communications. This was the main office. It was the heart of the system's communication line, and as such, it was well manned. Orders for everything emanated from Venus Equilateral. It was a delicate proposition, Venus Equilateral was, and hence the present-on-all-occasions official capacities and office staff.

This was the organization that Don Channing hoped to direct. A closed corporation with one purpose in mind, interplanetary communication!

Channing wondered if the summons for Walt Franks was an official one. Returning to the electronics office, Don punched the communicator and asked: "Is Walt in there?"

Arden's voice came back: "No, but Burbank is in Franks' office. Wanna listen?"

"Eavesdropper! Using the communicator?"

"Sure."

"Better shut it off," warned Don. "Burbank isn't foolish, you know, and there are pilot lights and warning flags on those things to tell if someone has the key open. I wouldn't want to see you fired for listening-in."

"All right, but it was getting interesting."

"If I'm betting on the right horse," said Channing, "this will be interesting for all before it is finished!"

Seven days went by in monotonous procession. Seven days in a world of constant climate. One week marked only by the changing of work shifts and the clocks that marked off the eight-hour periods. Seven days unmarred by rain or cold or heat. Seven days of uninterrupted sunshine that flickered in and out of the sealed viewports with eye-searing brilliance, coming and going as the station rotated.

But in the front offices, things were not as serene. No monotony to become irksome. Not that monotony ever set in seriously in the engineering department, but that sacred sanctum of all-things-that-didn't-behave-as-they-should found that even their usual turmoil was worse. There was nothing that a person could set his finger on directly. It was more of a quiet, undercover nature. On Monday Burbank sent around a communique removing the option of free messages for the personnel. On Tuesday he remanded the years-long custom of permitting the supply ships to carry, free, packages from friends at home. On Wednesday, Francis Burbank decided that there should be a curfew on the one and only beer emporium. That was made after he found that curtailing all sorts of alcoholics might easily lead to a more moral problem; there being little enough to do with one's spare time. On Thursday, he set up a stiff-necked staff of censors for the moving picture house. On Friday, he put a tax on cigarettes and candy. On Saturday, he installed time clocks in all the laboratories and professional offices, where previous to his coming, men had come for work a half hour late and worked an hour overtime at night.

On Sunday, he ran into trouble!

Don Channing stormed into the Director's office with a scowl on his face.

"Look," he said, "for years and years we have felt that any man, woman, or child that was willing to come out here was worth all the freedom and consideration that we could give them. What about this damned tax on

cigarettes? And candy? And who told you to stop our folks from telling their folks that they are still in good health? And why stop them from sending packages of candy, cake, mementoes, clothing, soap, mosquito dope, liquor, or anything else? Why shut off our beer half the day? Did you ever think that a curfew is something that can be applied only when time is one and the same for all? On Venus Equilateral, Mr. Burbank, six o'clock in the evening is two hours after dinner for one group, two hours after going to work for the second group, and mid-sleep for the third. Then this matter of cutting all love scenes, drinking, female vampires, banditry, bedroom items, murders, and sweater girls out of the movies? We are a selected group and well prepared to take care of our morality. Any man or woman going offside would be heaved out quick. Why, after years of personal freedom, do we find ourselves under the authority of a veritable dictatorship?"

Francis Burbank was not touched. "I'll trouble you to keep to your own laboratory," he told Channing. "Perhaps your own laxity in matters of this sort is the reason why the Commission preferred someone better prepared. You speak of many things. There will be more to come. I'll answer some of your questions. Why should we permit our profits to be eaten up by people sending messages, cost-free, to their acquaintances all over the minor planets? Why should valuable space for valuable supplies be taken up with personal favors between friends? And if the personnel wants to smoke and drink, let them pay for the privilege! It will help to pay for the high price of shipping the useless items out from the nearest planet—as well as saving of precious storage space!"

"But you're breeding ill will among the employees," objected Channing.

"Any that prefer to do so may leave!" snapped Burbank.

"You may find it difficult to hire people to spend their lives in a place that offers no sight of a sky or a breath of fresh air. The people here may go home to their own planets to find that the smell of fresh, spring air is more desirable than a climate that never varies from the personal optimum. I wonder, occasionally, if it might not be possible to instigate some sort of cold snap or a rainy season just for the purpose of bringing to the members of Venus Equilateral some of the surprises that are to be found in Chicago or New York. Hell, even Canalopolis has an occasional rainstorm!"

"Return to your laboratory," said Burbank coldly. "And let me run the station. Why should we spend useless money to pamper people? I don't care if Canalopolis does have an occasional storm, we are not on Mars, we are in Venus Equilateral. You tend to your end of the business and I'll do as I deem fitting for the station!"

Channing mentally threw up his hands and literally stalked out of the office. Here was a close-knit organization being shot full of holes by a screwball. He stamped down to the ninth level and beat upon the closed door of Joe's. The door remained closed.

Channing beat with his knuckles until they bled. Finally a door popped open down the hallway fifty yards and a man looked out. His head popped in again, and within thirty seconds the door to Joe's opened and admitted Channing.

Joe slapped the door shut behind Channing quickly.

"Whatinhell are you operating, Joe—a speakeasy?"

"The next time you want in," Joe informed him, "knock on 902 twice, 914 once, and then here four times. We'll let you in. And now, don't say anything too loud." Joe put a finger to his lips and winked broadly. "Even the walls listen," he said in a stage whisper.

He led Channing into the room and put on the light. There was a flurry of people who tried to hide their glasses under the table. "Never mind," called Joe. "It's only Dr. Channing."

The room relaxed.

"I want something stiff," Channing told Joe. "I've just gone three rounds with His Nibs and came out cold."

Some people within earshot asked about it. Channing explained what had transpired. The people seemed satisfied that Channing had done his best for them. The room relaxed into routine.

The signal knock came on the door and was opened to admit Walt Franks and Arden Westland. Franks looked as though he had been given a stiff workout in a cement mixer.

"Scotch," said Arden. "And a glass of brew for the lady."

"What happened to him?"

"He's been trying to keep to Burbank's latest suggestions."

"You've been working too hard," Channing chided him gently. "This is the wrong time to mention it, I suppose, but did that beam slippage have anything to do with your condition—or was it vice versa?"

"You know that I haven't anything to do with the beam controls personally," said Franks. He straightened up and faced Channing defiantly.

"Don't get mad. What was it?"

"Mastermind, up there, called me in to see if there were some manner or means of tightening the beam. I told him, sure, we could hold the beam to practically nothing. He asked me why we didn't hold the beam to a parallel and save the dispersed power. He claimed that we could reduce power by two to one if more of it came into the station instead of being smeared all over the firmament. I, foolishly, agreed with him. He's right. You could. But only if everything is immobilized. I've been trying to work out some means of controlling the beam magnetically so that it would compensate for the normal variations due to magnetic influences. So far I've failed."

"It can't be done. I know, because I worked on the problem for three years with some of the best brains in the system. To date, it is impossible."

A click attracted their attention. It was the pneumatic tube. A cylinder dropped out of the tube, and Joe opened it and handed the inclosed paper to Franks.

He read:

Walt: I'm sending this to you at Joe's because I know that is where you are and I think that you should get this real quick.

Helen S.

Walt smiled wearily and said: "A good secretary is a thing of beauty. A thing of beauty is admired and is a joy forever. Helen is both. She is a jewel."

"Yeah, we know. What does the letter say?"

"It is another communique from our doting boss. He is removing from my control the odd three hundred men I've got working on Beam Control. He is to assume the responsibility for them himself. I'm practically out of a job!"

"Make that two Scotches," Channing told Joe.

"Make it three," chimed in Arden. "I've got to work for him, too!"

"Is that so bad?" asked Channing. "All you've got to do is to listen carefully and do as you're told. We have to answer to the bird, too."

"Yeah," said Arden, "but you fellows don't have to listen to a dopey guy ask foolish questions all day. It's driving me silly."

"What I'd like to know," murmured Franks, "is what is the idea of pulling me off the job? Nuts, I've been on Beam Control for years. I've got the finest crew of men anywhere. They can actually foresee a shift and compensate for it, I think. I picked 'em myself and I've been proud of my outfit. Now," he said brokenly, "I've got no outfit. In fact, I have darned little crew left at all. Only my dozen lab members. I'll have to go back to swinging a meter myself before this is over."

It was quite a comedown. From the master of over three hundred highly paid, highly prized, intelligent technicians, Walt Franks was now the superintendent of one dozen laboratory technicians. It was a definite cut in his status with Communications.

Channing finished his drink and, seeing that Franks' attention was elsewhere, he told Arden: "Thanks for taking care of him, but don't use all your sympathy on him. I feel that I'm going to need your shoulder to cry on before long."

"Any time you want a soft shoulder," said Arden generously, "let me know. I'll come a-running."

Channing went out. He roamed nervously all the rest of the day. He visited the bar several times, but the general air of the place depressed him. From a place of recreation, laughter, and pleasantries, Joe's place had changed to a room for reminiscences and remorse, a place to drown one's troubles—or poison them—or to preserve them in alcohol.

He went to see the local moving picture, a piece advertised as being one of the best mystery thrillers since DeMille. He found that all of the interesting parts were cut out and that the only thing that remained was a rather disjointed portrayal of a portly detective finding meaningless clues and ultimately the criminal. There was a suggestion at the end, that the detective and the criminal had fought it out, but whether it was with pistols, field pieces, knives, cream puffs, or words was left up to the imagination. It was also to be assumed that he and the heroine, who went into a partial blackout every time she sat down, finally got acquainted enough to hold hands after the picture.

Channing stormed out of the theater after seeing the above and finding that the only cartoon had been barred because it showed an innocuous cow without benefit of shorts.

He troubled Joe for a bottle of the best and took to his apartment in disappointment. By eight o'clock in the evening, Don Channing was asleep with all of his clothing on. The bed rolled and refused to stay on an even keel, but Channing found a necktie and tied himself securely in the bed and died off in a beautiful, boiled cloud.

He awoke to the tune of a beautiful hangover. He gulped seven glasses of water and staggered to the shower. Fifteen minutes of iced needles, and some coffee brought him part way back to his own, cheerful self. He headed down the hall toward the elevator.

He found a note in his office directing him to appear at a conference in Burbank's office. Groaning in anguish, Don went to the Director's office expecting the worst.

It was bad. In fact, it was enough to drive everyone in the conference to drink. Burbank asked opinions on everything, and then tore the opinions apart with little regard to their validity. He expressed his own opinion many times, which was a disgusted sense of the personnel's inability to do anything of real value.

"Certainly," he stormed, "I know you are operating. But have there been any new developments coming out of your laboratory, Mr. Channing?"

Someone was about to tell Burbank that Channing had a doctor's degree, but Don shook his head.

"We've been working on a lot of small items," said Channing. "I cannot say whether there has been any one big thing that we could point to. As we make developments, we put them into service. Added together, they make quite an honest effort."

"What, for instance?" stormed Burbank.

"The last one was the coupler machine improvement that permitted better than a thousand words per minute."

"Up to that time the best wordage was something like eight hundred words," said Burbank. "I think that you have been resting too long on your laurels. Unless you can bring me something big enough to advertise, I shall have to take measures."

"Now you, Mr. Warren," continued Burbank. "You are the man who is supposed to be superintendent of maintenance. May I ask why the outer hull is not painted?"

"Because it would be a waste of paint," said Warren. "Figure out the acreage of a surface of a cylinder three miles long and a mile in diameter. It is almost eleven square miles! Eleven square miles to paint from scaffolding hung from the outside itself."

"Use bos'ns' chairs," snapped Burbank.

"A bos'n's chair would be worthless," Warren informed Burbank. "You must remember that to anyone trying to operate on the outer hull, the outer hull is a ceiling and directly overhead."

"Another thing," said Warren, "you paint that hull and you'll run this station by yourself. Why d'ye think we have it shiny?"

"If we paint the hull," persisted Burbank, "it will be more presentable than that nondescript steel color."

"That steel color is as shiny as we could make it," growled Warren. "We want to get rid of as much radiated heat as we can. You slap a coat of any kind of paint on that hull and you'll have plenty of heat in here."

"Ah, that sounds interesting. We'll save heating costs—"

"You idiot," snapped Warren. "You fool. Sure we'll have heat in here. We'll save some heating costs. But do you realize that we'd have no opportunity to control it? We're on a safe margin now. We radiate just a little more than we receive. We make up the rest by artificial heating. But there have been occasions when it became necessary to dissipate a lot of energy in here for one reason or another, and then we've had to shut off the fires. What would happen if we couldn't cool the damned coffee can? We'd roast the first time that we got a new employee with a body temperature a half degree above normal!"

"You're being openly rebellious," Burbank warned him.

"So I am. And if you persist in your attempt to make this place presentable, you'll find me and my gang outright mutinous! Good day, sir!"

He stormed out of the office and slammed the door.

"Take a note, Miss Westland. Interplanetary Communications Commission, Terra. Gentlemen: Michael Warren, superintendent of maintenance at Venus Equilateral, has proven to be unreceptive to certain suggestions as to the appearance and/or operation of Venus Equilateral. It is my request that he be replaced immediately. Signed, Francis Burbank, Director." He paused to see what effect that message had upon the faces of the men around the table. "Send that by special delivery!"

Johnny Billings opened his mouth to say something, but shut it with a snap. Westland looked up at Burbank, but she said nothing. Arden gave Channing a sly smile, and Channing smiled back. There were grins about the table, too, for everyone recognized the boner. Burbank had just sent a letter from the interworld communications relay station by special delivery *mail*. It

would not get to Terra for better than two weeks; a use of the station's facilities would have the message in the hands of the Commission within the hour.

"That will be all, gentlemen." Burbank smiled smugly. "Our next conference will be next Monday morning!"

"Mr. Channing," chortled the pleasant voice of Arden Westland. "Now that the trifling influence of the boss versus secretary taboo is off, will you have the pleasure of buying me a drink?"

"Can you repeat that word for word and explain it?" grinned Don.

"A man isn't supposed to make eyes at his secretary. A gal ain't supposed to seduce her boss. Now that you are no longer Acting Director, and I no longer your stenog, how about some sociability?"

"I never thought that I'd be propositioned by a typewriter jockey," said Channing, "but I'll do it. What time is it? Do we do it openly, or must we sneak over to the apartment and snaffle a snort on the sly?"

"We snaffle. That is, if you trust me in your apartment."

"I'm scared to death," Channing informed her. "But if I should fail to defend my honor, we must remember that it is no true failure if I try and fail!"

"That sounds like a nice alibi," said Arden with a smile. "Or a come-on. I don't know which. Or, Mr. Channing, am I being told that my advances might not be welcome?"

"We shall see," Channing said. "We'll have to make a careful study of the matter. I cannot make any statements without first making a thorough investigation under all sorts of conditions. Here we are. You will precede me through the door, please."

"Why?" asked Arden.

"So that you cannot back out at the last possible moment. Once I get you inside, I'll think about keeping you there!"

"As long as you have some illegal fluid, I'll stay." She tried to leer at Don but failed because she had had all too little experience in leering. "Bring it on!"

"Here's to the good old days," toasted Don as the drinks were raised.

"Nope. Here's to the future," proposed Arden. "Those good old days—all they were was old. If you were back in them, you'd still have to have the pleasure of meeting Burbank."

"Grrrrr," growled Channing. "That name is never mentioned in this household."

"You haven't a pix of the old bird turned to the wall, have you?" asked Arden.

"I tossed it out."

"We'll drink to that." They drained glasses. "And we'll have another!"

"I need another," said Channing. "Can you imagine that buzzard asking me to invent something big in seven days?"

"Sure. By the same reasoning that he uses to send a letter from Equilateral Station instead of just slipping it in on the Terra beam. Faulty."

"Phony."

The door opened abruptly and Walt Franks entered. "D'ja hear the latest?" he asked breathlessly.

"No," said Channing. He was reaching for another glass automatically. He poured, and Walt watched the amber fluid creep up the glass, led by a sheet of white foam.

"Then look!" Walt handed Channing an official envelope. It was a regular notice to the effect that there had been eleven failures of service through Venus Equilateral.

"Eleven! What makes?"

"Mastermind."

"What's he done?"

"Remember the removal of my jurisdiction over the beam control operators? Well, in the last ten days, Burbank has installed some new features to cut expenses. I think that he hopes to lay off a couple of hundred men."

"What's he doing, do you know?"

"He's shortening the dispersion. He intends to cut the power by slamming more of the widespread beam into the receptor. The tighter beam makes aiming more difficult, you know, because at seventy million miles, every time little Joey on Mars swings his toy horseshoe magnet on the end of his string, the beam wabbles. And at seventy million miles, how much wobbling does it take to send a narrow beam clear off the target?"

"The normal dispersion of the beam from Venus is over a thousand miles wide. It gyrates and wabbles through most of that arc. That is why we picked that particular dispersion. If we could have pointed the thing like an arrow, we'd have kept the dispersion down."

"Right. And he's tightened the beam to less than a hundred miles dispersion. Now, every time that a sunspot gets hit amidships with a lady sunspot, the beam goes off on a tangent. We've lost the beam eleven times in a week. That's more times than I've lost it in three years!"

"O.K.," said Channing. "So what? Mastermind is responsible. We'll sit tight and wait for developments. In any display of abilities, we can spike Mr. Burbank. Have another drink?"

"Got any more? If you're out, I've got a couple of cases cached underneath the bed in my apartment."

"I've plenty," said Channing. "And I'll need plenty. I have exactly twenty-two hours left in which to produce something comparable to the telephone, the electric light, the airplane, or the expanding Universe! Phooey. Pour me another, Arden."

A knock at the door; a feminine voice interrupted simultaneously. "May I come in?"

It was Walt's secretary. She looked worried. In one hand she waved another letter.

"Another communique?" asked Channing.

"Worse. Notice that for the last three hours, there have been less than twelve percent of messages relayed!"

"Five minutes' operation out of an hour," said Channing. "Where's that from?"

"Came out on the Terra beam. It's marked number seventeen, so I guess that sixteen other tries have been made."

"What has Mastermind tried this time?" stormed Channing. He tore out of the room and headed for the Director's office on a dead run. On the way, he hit his shoulder on the door, caromed off the opposite wall, righted himself, and was gone in a flurry of flying feet. Three heads popped out of doors to see who was making the noise.

Channing skidded into Burbank's office on his heels. "What gives?" he snapped. "D'ye realize that we've lost the beam? What have you been doing?"

"It is a minor difficulty," said Burbank calmly. "We will iron it out presently."

"Presently! Our charter doesn't permit interruptions of service of this magnitude. I ask again: What are you doing?"

"You, as electronics engineer, have no right to question me. I repeat, we shall iron out the difficulty presently."

Channing snorted and tore out of Burbank's office. He headed for the Office of Beam Control, turned the corner on one foot, and slammed the door in roughly.

"Chuck!" he yelled. "Chuck Thomas! Where are you?"

No answer. Channing left the beam office and headed for the master control panels, out near the air lock end of Venus Equilateral. He found Thomas stewing over a complicated piece of apparatus.

"Chuck, for the love of Michael, what in the devil is going on?"

"Thought you knew," answered Thomas. "Burbank had the crew install photoelectric mosaic banks. Instead of having a crew of beam-control operators, he intends to use the photomosaics to keep Venus, Terra, and Mars on the beam."

"Great sniveling Scott. They tried that in the last century and tossed it out three days later. Where's the crew now?"

"Packing for home. They've been laid off!"

"Get 'em back! Put 'em to work. Turn off those darned photomosaics and use the manual again. We've lost every beam we ever had."

A sarcastic voice came in at this point. "For what reason do you interfere with my improvements?" sneered the voice. "Could it be that you are accepting graft from the employees to keep them on the job by preventing the installation of superior equipment?"

Channing turned on his toe and let Burbank have one. It was a neat job, coming up at the right time and connecting sweetly. Burbank went over on his head.

"Get going," Channing snapped at Thomas.

Charles Thomas was not a small man himself, but after considering Channing's one ninety, he decided to comply. He left.

Channing shook Burbank's shoulder. He slapped the man's face. Eyes opened; accusing eyes, rendered mute by a very sore jaw, tongue, and throat.

"Now listen," snapped Channing. "Listen to every word! Mosaic directors are useless. Have you any reason why? It is because of the lag. At planetary distances, light takes an appreciable time to reach your planet. The beam wabbles, swerves out of line because of intervening factors; varying magnetic fields, even the bending of light due to gravitational fields will make the beam swerve microscopically. But, Burbank, a microscopic discrepancy is all that is needed to bust things wide open. You've got to have experienced men to operate the beam controls. Men who can think.

Men who can, from experience, reason that this fluctuation will not last, but will swing back in a few seconds, or that this type of swerving will increase in magnitude for a half-hour, maintain the status, and then return, pass through zero and find the same level on the minus side.

"Since light and centimeter waves are not exactly alike in performance, a field that will swerve one may not affect the other. Ergo your photomosaic is useless. The photoelectric mosaic is a brilliant gadget for keeping a plane in a spotlight or for aiming a sixteen-inch gun, but it is worthless for anything over a couple of million miles.

"So I've called the men back to their stations. And don't try anything foolish again without consulting the men who are paid to think!"

Channing got up and left. As he strode down the stairs to the apartment level, he met many of the men who had been laid off. None of them said a word, but all of them wore bright, knowing smiles.

Monday morning! Burbank was himself again. The rebuff given him by Don Channing had worn off and he was sparkling with ideas. He speared Franks with the glitter in his eye and said: "If our beams are always on the center, why is it necessary to use multiplex-diversity?"

Franks smiled. "You're mistaken," he told Burbank. "They're not always on the button. They vary. Therefore we use diversity transmission so that if one beam fails momentarily, one of the other beams will bring the signal in. It is analogous to tying five or six ropes onto a hoisted stone. If one breaks, you have the others."

"You have them running all the time, then?"

"Certainly. At several minutes of time-lag in transmission, to try and establish a beam failure of a few seconds' duration is utter foolishness."

"And you disperse the beam to a thousand miles wide to keep the beam centered at any variation?" Burbank shot at Channing.

"Not for any variation. Make that any *normal* gyration and I'll buy it."

"Then why don't we disperse the beam to two or three thousand miles and do away with diversity transmission?" asked Burbank triumphantly.

"Ever heard of fading?" asked Channing with a grin. "Your signal comes and goes. Not gyration, it just gets weaker. It fails for want of something to eat, I guess, and takes off after a wandering cosmic ray. At any rate, there are many times per minute that one beam will be right on the nose and yet so weak that our strippers cannot clean it enough to make it usable. Then the diversity system comes in handy. Our coupling detectors automatically select the proper signal channel. It takes the one that is the strongest and subdues the rest within itself."

"Complicated?"

"It was done in the heyday of radio—1935 or so. Your two channels come in to a common detector. Automatic volume control voltage comes from the single detector and is applied to all channels. This voltage is proper for the strongest channel, but is too high for the ones receiving the weaker signal, blocking them by rendering them insensitive. When the strong channel fades and the weak channel rises, the detector follows down until the two signal channels are equal and then it rises with the stronger channel."

"I see," said Burbank. "Has anything been done about fading?"

"It is like the weather, according to Mark Twain," smiled Channing. "'Everybody talks about it, but nobody does anything about it.' About all we've learned is that we can cuss it out and it doesn't cuss back."

"I think it should be tried," said Burbank.

"If you'll pardon me, it has been tried. The first installation at Venus Equilateral was made that way. It didn't work, though we used more power than all of our diversity transmitter together. Sorry."

"Have you anything to report?" Burbank asked Channing.

"Nothing. I've been more than busy investigating the trouble we've had in keeping the beams centered."

Burbank said nothing. He was stopped. He hoped that the secret of his failure was not generally known, but he knew at the same time that when three hundred men are aware of something interesting, some of them will see to it that all others involved will surely know. He looked at the faces of

the men around the table and saw suppressed mirth in every one of them. Burbank writhed in inward anger. He was a good poker player. He didn't show it at all.

He then went on to other problems. He ironed some out, others he shelved for the time being. Burbank was a good businessman, give him credit. But like so many other businessmen, Burbank had the firm conviction that if he had the time to spare and at the same time was free of the worries and paper work of his position, he could step into the laboratory and show the engineers how to make things hum. He was infuriated every time he saw one of the engineering staff sitting with hands behind head, lost in a gazy, unreal land of deep thought. Though he knew better, he was often tempted to raise hell because the man was obviously loafing.

But give him credit. He could handle business angles to perfection. In spite of his tangle over the beam control, he had rebounded excellently and had ironed out all of the complaints that had poured in. Ironed it out to the satisfaction of the injured party as well as the Interplanetary Communications Commission, who were interested in anything that cost money.

He dismissed the conference and went to thinking. And he assumed the same pose that infuriated him in other men under him; hands behind head, feet upon desk.

The moving picture theater was dark. The hero reached longing arms to the heroine, and there was a sort of magnetic attraction. They approached one another. But the spark misfired. It was blacked out with a nice slice of utter blackness that came from the screen and spread its lightlessness all over the theater. In the ensuing darkness, there were several osculations that were more personal and more satisfying than the censored clinch. The lights flashed on and several male heads moved back hastily. Female lips smiled happily. Some of them parted in speech.

One of them said: "Why, Mr. Channing!"

"Shut up, Arden," snapped the man. "People will think that I've been kissing you."

"If someone else was taking advantage of the situation," she said, "you got gypped. I thought I was kissing you and I cooked with gas!"

"Did you ever try that before?" asked Channing interestedly.

"Why?" she asked.

"I liked it. I merely wondered, if you'd worked it on other men, what there was about you that kept you single."

"They all died after the first application," she said. "They couldn't take it."

"Let me outta here! I get the implication. I am the first bird that hasn't died, hey?" He yawned luxuriously.

"Company or the hour?" asked Arden.

"Can't be either," he said. "Come on, let's break a bottle of beer open. I'm dry!"

"I've got a slight headache," she told him. "From what, I can't imagine."

"I haven't a headache, but I'm sort of logy."

"What have you been doing?" asked Arden. "Haven't seen you for a couple of days."

"Nothing worth mentioning. Had an idea a couple of days ago and went to work on it."

"Haven't been working overtime or missing breakfast?"

"Nope."

"Then I don't see why you should be ill. I can explain my headache away by attributing it to eyestrain. Since Billyboy came here, and censored the movies to the bone, the darned things flicker like anything. But eyestrain doesn't create an autointoxication. So, my fine fellow, what have you been drinking?"

"Nothing that I haven't been drinking since I first took to my second bottlehood some years ago."

"You wouldn't be suffering from a hangover from that hangover you had a couple of weeks ago?"

"Nope. I swore off. Never again will I try to drink a whole quart of Two Moons in one evening. It got me."

"It had you for a couple of days," laughed Arden. "All to itself."

Don Channing said nothing. He recalled, all too vividly, the rolling of the tummy that ensued after that session with the only fighter that hadn't yet been beaten: Old John Barleycorn.

"How are you coming on with Burbank?" asked Arden. "I haven't heard a rave for—well, ever since Monday morning's conference. Three days without a nasty dig at Our Boss. That's a record."

"Give the devil his due. He's been more than busy placating irate citizens. That last debacle with the beam control gave him a real Moscow winter. His reforms came to a stop whilst he retrenched. But he's been doing an excellent job of squirming out from under. Of course, it has been helped by the fact that even though the service was rotten for a few hours, the customers couldn't rush out to some other agency to get communications with the other planets."

"Sort of: 'Take us, as lousy as we are?'"

"That's it."

Channing opened the door to his apartment and Arden went in. Channing followed, and then stopped cold.

"Great Jeepers!" he said in an awed tone. "If I didn't know—"

"Why, Don! What's so startling?"

"Have you noticed?" he asked. "It smells like the inside of a chicken coop in here!"

Arden sniffed. "It does sort of remind me of something that died and couldn't get out of its skin." Arden smiled. "I'll hold my breath. Any sacrifice for a drink."

"That isn't the point. This is purified air. It should be as sweet as a baby's breath."

"Some baby," whistled Arden. "What's she been drinking?"

"It wasn't cow juice. Nor yet lake filler. What I've been trying to put over is that the air doesn't seem to have been changed in here for nine weeks."

Channing went to the ventilator and lit a match. The flame bent over, flickered, and went out.

"Air intake is O.K.," he said. "Maybe it is I. Bring on that bottle, Channing; don't keep the lady waiting."

He yawned again, deeply and jaw-stretchingly. Arden yawned too, and the thought of both of them stretching their jaws to the breaking-off point made both of them laugh foolishly.

"Arden, I'm going to break one bottle of beer with you, after which I'm going to take you home, kiss you goodnight, and toss you into your own apartment. Then I'm coming back here and I'm going to hit the hay!"

Arden took a long, deep breath. "I'll buy that," she said. "And tonight, it wouldn't take much persuasion to induce me to snooze right here in this chair!"

"Oh fine," cheered Don. "That would fix me up swell with the neighbors. I'm not going to get shotgunned into anything like that!"

"Don't be silly," said Arden.

"From the look in your eye," said Channing, "I'd say that you were just about to do that very thing. I was merely trying to dissolve any ideas that you might have."

"Don't bother," she said pettishly. "I haven't any ideas. I'm as free as you are, and I intend to stay that way!"

Channing stood up. "The next thing we know, we'll be fighting," he observed. "Stand up, Arden. Shake."

Arden stood up, shook herself, and then looked at Channing with a strange light in her eyes. "I feel sort of dizzy," she admitted. "And everything irritates me."

She passed a hand over her eyes wearily. Then, with a visible effort, she straightened. She seemed to throw off her momentary ill feeling instantly. She smiled at Channing and was her normal self in less than a minute.

"What is it?" she asked. "Do you feel funny, too?"

"I do!" he said. "I don't want that beer. I want to snooze."

"When Channing would prefer snoozing to boozing, he is sick," she said. "Come on, fellow, take me home."

Slowly they walked down the long hallway. They said nothing. Arm in arm they went, and when they reached Arden's door, their goodnight kiss lacked enthusiasm. "See you in the morning," said Don.

Arden looked at him. "That mug was a little flat. We'll try it again—tomorrow or next week."

Don Channing's night sleep was broken by dreams. He was warm. His dreams depicted him in a warm, airless chamber, and he was forced to breathe that same stale air again and again. He awoke in a hot sweat, weak and feeling—well, lousy!

He dressed carelessly. He shaved hit-or-miss. His morning coffee tasted flat and sour. He left the apartment in a bad mood, and bumped into Arden at the corner of the hall.

"Hello," she said. "I feel rotten. But you have improved. Or is that passionate breathing just a lack of fresh air?"

"Hell! That's it!" he said. He snapped up his wrist watch, which was a chronograph, equipped with a stop-watch hand. He looked about, and finding a man sitting on a bench, apparently taking it easy while waiting for someone, Channing clicked the sweep hand into gear. He started to count the man's respiration.

"What gives?" asked Arden. "What's 'It'? Why are you so excited? Did I say something?"

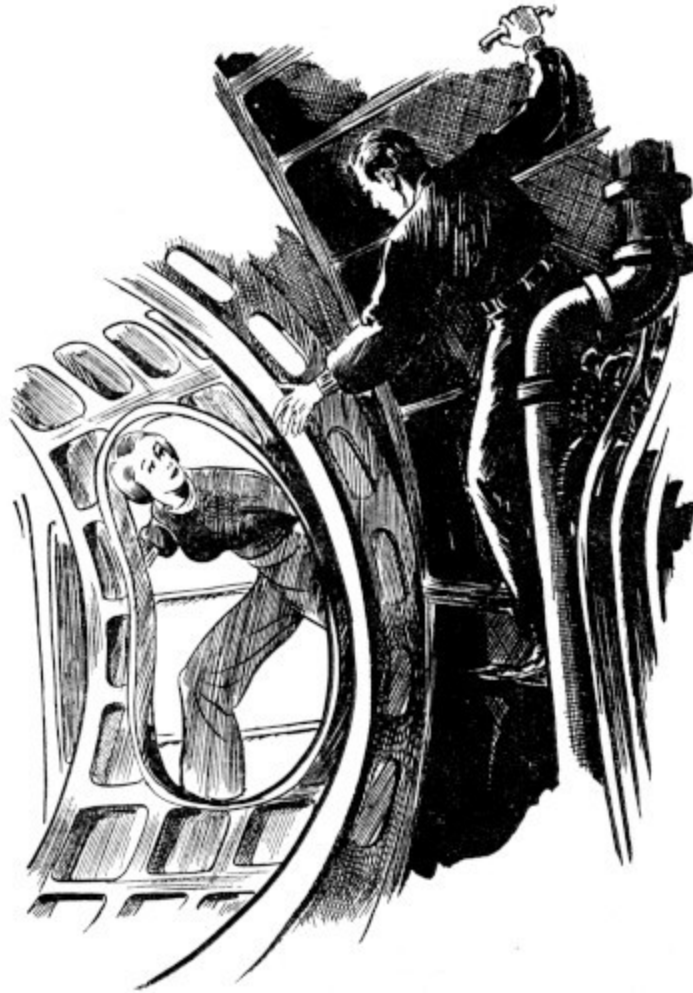
"You did," said Channing after fifteen seconds. "That bird's respiration is better than fifty! This whole place is filled to the gills with carbon dioxide. Come on, Arden, let's get going!"

Channing led the girl by several yards by the time that they were within sight of the elevator. He waited for her, and then sent the car upward at a full throttle. Minutes passed, and they could feel that stomach-rising sensation that comes when gravity is lessened. Arden clasped her hands over her middle and hugged. She squirmed and giggled.

"You've been up to the axis before," said Channing. "Take long, deep breaths."

The car came to a stop with a slowing effect. A normal braking stop would have catapulted them against the ceiling. "Come on," he grinned at her, "here's where we make time!"

Channing looked up the little flight of stairs that led to the innermost level. He winked at Arden and jumped. He passed up through the opening easily. "Jump," he commanded. "Don't use the stairs!"



Arden jumped. She sailed upward, and as she passed through the opening, Channing caught her by one arm and stopped her flight. "At that speed you'd go right on across," he said.

She looked up, and there about two hundred feet overhead she could see the opposite wall.

Channing snapped on the lights. They were in a room two hundred feet in diameter and three hundred feet long. "We're at the center of the station," Channing informed her. "Beyond that bulkhead is the air lock. On the other side of the other bulkhead, we have the air plants, the storage spaces, and several cubic inches of machinery."

"Inches?" asked Arden. Then she saw that he was fooling.

"Come on," he said. He took her by the hand and with a kick he propelled himself along on a long, curving course to the opposite side of the inner cylinder. He gained the opposite bulkhead as well.

"Now, that's what I call traveling," said Arden. "But my tummy goes *whoosh, whoosh*, every time we cross the center."

Channing operated a heavy door. They went in through rooms full of machinery and into rooms stacked to the center with boxes; stacked from the wall to the center and then packed with springs. Near the axis of the cylinder, things weighed so little that packing was necessary to keep them from bouncing around.

"I feel giddy," said Arden.

"High in oxygen," said he. "The CO₂ drops to the bottom, being heavier. Then, too, the air is thinner up here because centrifugal force swings the whole out to the rim. Out there we are so used to 'down' that here, a half mile above—or to the center, rather—we have trouble in saying, technically, what we mean. Watch!"

He left Arden standing and walked rapidly around the inside of the cylinder. Soon he was standing on the steel plates directly above her head. She looked up, and shook her head.

"I know why," she called, "but it still makes me dizzy. Come down from up there. Or I'll be sick."

Channing made a neat dive from his position above her head. He did it merely by jumping upward from his place toward her place, apparently hanging head down from the ceiling. He turned a neat flip-flop in the air and landed easily beside her. Immediately, for both of them, things became right-side-up again.

Channing opened the door to the room marked: "Air Plant." He stepped in, snapped on the lights, and gasped in amazement. The room was empty! Completely empty! Absolutely, and irrevocably vacant. Oh, there was some

dirt on the floor and some trash in the corners, and a trail of scratches on the floor to show that the life-giving air plant had been removed, hunk by hunk, out through another door at the far end of the room.

"Whoa, Tillie!" screamed Don. "We've been stabbed! Arden, get on the - type and have ... no, wait a minute until we find out a few more things about this!"

They made record time back to the office level. They found Burbank in his office, leaning back, and talking to someone on the phone.

Channing tried to interrupt, but Burbank removed his nose from the telephone long enough to snarl, "Can't you see I'm busy? Have you no manners or respect?"

Channing, fuming inside, swore inwardly. He sat down with a show of being calm and folded his hands over his abdomen like the famed statue of Buddha. Arden looked at him, and for all of the trouble they were in, she couldn't help giggling. Channing, tall, lanky, and yet strong, looked as little as possible like the popular, pudgy figure of the Sitting Buddha.

A minute passed.

Burbank hung up the phone.

"Where does Venus Equilateral get its air from?" snapped Burbank.

"That's what I want—"

"Answer me, please. I'm worried."

"So am I. Something—"

"Tell me first, from what source does Venus Equilateral get its fresh air?"

"From the air plant. And that is—"

"There must be more than one," said Burbank thoughtfully.

"There's only one."

"There *must* be more than one. We couldn't live if there weren't," said the Director.

"Wishing won't make it so. There is only one."

"I tell you, there must be another. Why, I went into the one up at the axis yesterday and found that instead of a bunch of machinery, running smoothly, purifying air, and sending it out to the various parts of the station, all there was was a veritable jungle of weeds. Those weeds, Mr. Channing, looked as though they must have been put in there years ago. Now, where did the air-purifying machinery go?"

Channing listened to the latter half of Burbank's speech with his chin at half-mast. He looked as though a feather would knock him clear across the office.

"I had some workmen clear the weeds out. I intend to replace the air machinery as soon as I can get some new material sent from Terra."

Channing managed to blink. It was an effort. "You had workmen toss the weeds out—" he repeated dully. "The weeds—"

There was silence for a minute. Burbank studied the man in the chair as though Channing were a piece of statuary. Channing was just as motionless. "Channing, man, what ails you—" began Burbank. The sound of Burbank's voice aroused Channing from his shocked condition.

Channing leaped to his feet. He landed on his heels, spun, and snapped at Arden: "Get on the -type. Have 'em slap as many oxy-drums on the fastest ship they've got! Get 'em here at full throttle. Tell 'em to load up the pilot and crew with gravanol and not to spare the horsepower! Scram!"

Arden gasped. She fled from the office.

"Burbank, what did you think an air plant was?" snapped Channing.

"Why, isn't it some sort of purifying machinery?" asked the wondering Director.

"What better purifying machine is there than a plot of grass?" shouted Channing. "Weeds, grass, flowers, trees, alfalfa, wheat, or anything that grows and uses chlorophyll. We breathe oxygen, exhale CO₂. Plants inhale CO₂ and exude oxygen. An air plant means just that. It is a specialized type of Martian sawgrass that is more efficient than anything else in the system for inhaling dead air and revitalizing it. And you've tossed the weeds out!" Channing snorted in anger. "We've spent years getting that plant so that it will grow just right. It got so good that the CO₂ detectors weren't even

needed. The balance was so adjusted that they haven't even been turned for three or four years. They were just another source of unnecessary expense. Why, save for a monthly inspection, that room isn't even opened, so efficient is the Martian sawgrass. We, Burbank, are losing oxygen!"

The Director grew white. "I didn't know," he said.

"Well, you know now. Get on your horse and do something. At least, Burbank, stay out of my way while I do something."

"You have a free hand," said Burbank. His voice sounded beaten.

Channing left the office of the Director and headed for the chem lab. "How much potassium chlorate, nitrate, sulphate, and other oxygen-bearing compounds have you?" he asked. "That includes mercuric oxide, spare water, or anything else that will give us oxygen if broken down?"

There was a ten-minute wait until the members of the chem lab took a hurried inventory.

"Good," said Channing. "Start breaking it down. Collect all the oxygen you can in containers. This is the business! It has priority! Anything, no matter how valuable, must be scrapped if it can facilitate the gathering of oxygen. God knows, there isn't by half enough—not even a tenth. But try, anyway."

Channing headed out of the chemistry laboratory and into the electronics lab. "Jimmy," he shouted. "Get a couple of stone jars and get an electrolysis outfit running. Fling the hydrogen out of a convenient outlet into space and collect the oxygen. Water, I mean. Use tap water, right out of the faucet."

"Yeah, but—"

"Jimmy, if we don't breathe, what chance have we to go on drinking? I'll tell you when to stop."

"O.K., Doc," said Jimmy.

"And look. As soon as you get that running, set up a CO₂ indicator and let me know the percentage at the end of each hour! Get me?"

"I take it that something has happened to the air plant?"

"It isn't functioning," said Channing shortly. He left the puzzled Jimmy and headed for the beam-control room. Jimmy continued to wonder about the air plant. How in the devil could an air plant cease functioning unless it were—*dead*! Jimmy stopped wondering and began to operate on his electrolysis set-up furiously.

Channing found the men in the beam-control room worried and ill at ease. The fine co-ordination that made them expert in their line was ebbing. The nervous work, that made it necessary to run the men in ten-minute shifts with a half hour of rest in between, demanded perfect motor control, excellent perception, and a fine power of reasoning. The barely perceptible lack of oxygen at this high level was taking its toll already.

"Look, fellows, we're in a mess. Until further notice, take five-minute shifts. We've got about thirty hours to go. If the going gets tough, drop it to three-minute shifts. But, fellows, keep those beams centered until you drop!"

"We'll keep 'em going if we have to call our wives up here to run 'em for us," said one man. "What's up?"

"Air plant's sour. Losing oxy. Got a shipload coming out from Terra, be here in thirty hours. But upon you fellows will rest the responsibility of keeping us in touch with the rest of the system. If you fail, we could call for help until hell freezes us all in—and no one would hear us!"

"We'll keep 'em rolling," said a little fellow that had to sit on a tall stool to get even with the controls.

Channing looked out of the big plastiglass dome that covered the entire end of the Venus Equilateral Station. "Here messages go in and out," he mused. "The other end brings us things that take our breath away."

Channing was referring to the big air lock at the other end of the station, three miles away, right through the center.

At the center of the dome, there was a sighting 'scope. It kept Polaris on a marked circle, keeping the station exactly even with the Terrestrial North. About the periphery of the dome, looking out across space, the beam control operators were sitting, each with a hundred-foot parabolic reflector below his position, outside the dome, and under the rim of the transparent

bowl. These reflectors shot the interworld signals across space in tight beams, and the men, half the time anticipating the vagaries of space-warp, kept them centered on the proper, shining speck in that field of stars.

Above his head, the stars twinkled. Puny man, setting his will against the monstrous void. Puny man, dependent upon atmosphere. "Nature abhors a vacuum," once said Torrecelli. What braggadocio! If Nature abhorred a vacuum, why did she make so much of it?

Arden Westland entered the apartment without knocking. "I'd give my right arm up to here for a cigarette," she said, marking above the elbow with the edge of her other hand.

"Na-hah," said Channing. "Can't burn oxygen."

"I know. I'm tired, I'm cold, and I'm ill. Anything you can do for a lady?"

"Not as much as I'd like to do," said Channing. "I can't help much. We've got most of the place stopped off with the air-tight doors. We've been electrolyzing water, baking KClO_3 , and everything else we can get oxy out of. I've a crew of men trying to absorb the CO_2 . Jimmy Dickson is bringing me hourly reports on the CO_2 content and we are losing. Slowly, Arden, but we are losing. Of course, I've known all along that we couldn't support the station on the meager supplies we have on hand. But we'll win in the end. Our microcosmic world is getting a shot in the arm in a few hours that will reset the balance."

"I don't see why we didn't prepare for this emergency," said Arden.

"This station is well balanced. There are enough people here and enough space to make a little world of our own. We can establish a balance that is pretty darned close to perfect. The imperfections are taken care of by influxes of supplies from the system. Until Burbank upset the balance, we could go on forever, utilizing natural purification of air and water. We grow a few vegetables and have some meat critters to give milk and steak. The energy to operate Venus Equilateral is supplied with the photoelectric collectors—sun power, if you please. Why should we burden ourselves with

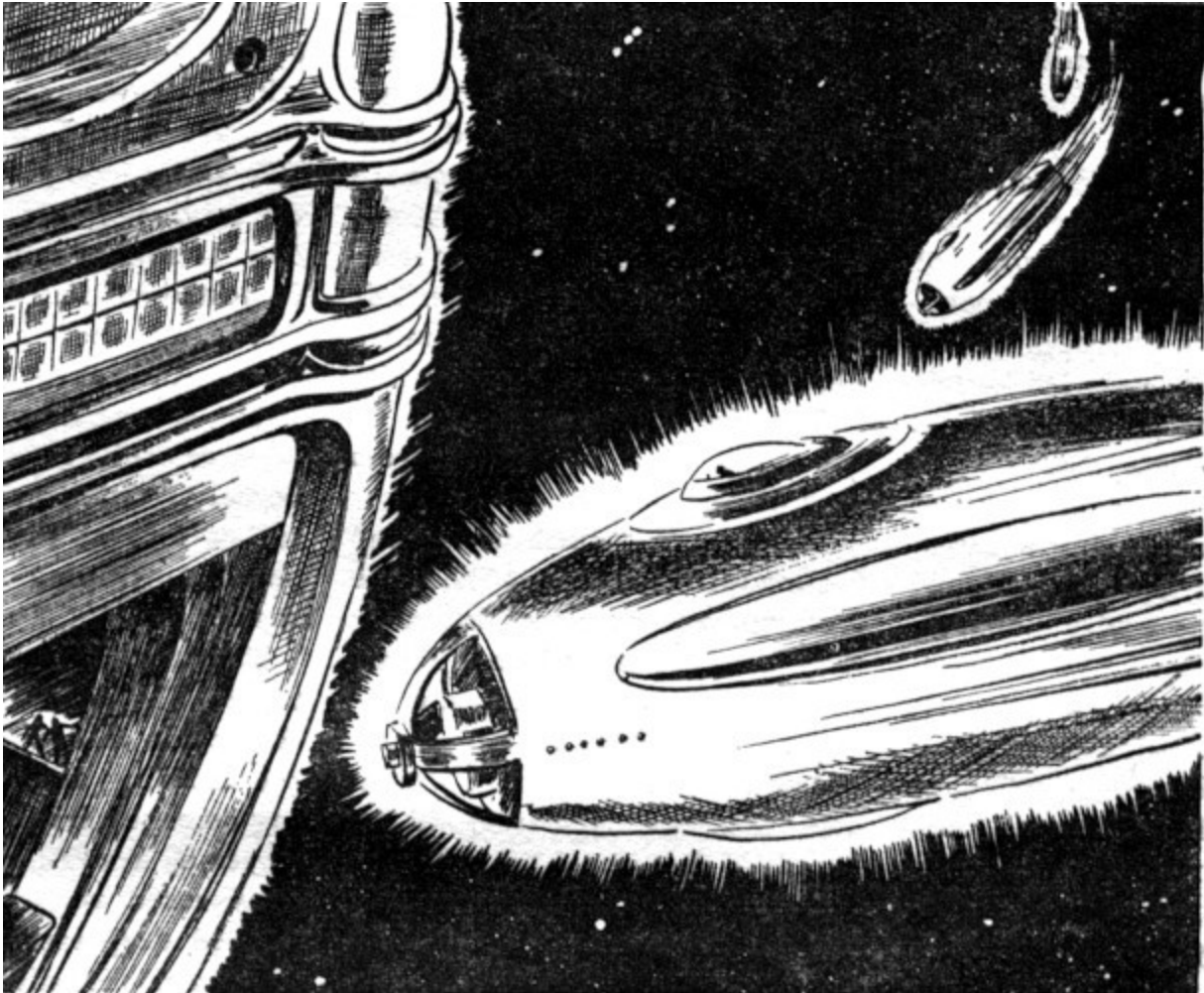
a lot of cubic feet of supplies that would take up room necessary to maintain our balance? We are not in bad shape. We'll live, though we'll all be a bunch of tired, irritable people who yawn in one another's faces."

"And after it is over?"

"We'll establish the balance. Then we'll settle down again. We can take up where we left off."

"Not quite. Venus Equilateral has been seared by fire. We'll be tougher and less tolerant of outsiders. If we were a closed corporation before, we'll be tighter than a vacuum-packed coffee can afterwards. And the first bird that cracks us will get hissed at."

Three superliners hove into sight at the end of thirty-one hours. They circled the station, signaling by helio. They approached the air lock end of the station and made contact. Their bulk tipped the station slightly, tipped it and rotated it by gyroscopic reaction. The air lock was opened and space-suited figures swarmed over the mile-wide end of the station. A stream of big oxygen tanks were brought into the air lock, admitted, and taken to the last bulwark of huddled people on the fourth level.



From one of the ships there came a horde of men carrying huge square trays of dirt and green, growing sawgrass.

For six hours, Venus Equilateral was the scene of wild, furious activity. The dead air was blown out of bad areas, and the hissing of oxygen tanks was heard in every room. Gradually the people left the fourth level and returned to their rightful places. The station rang with laughter once more, and business, stopped short for want of breath, took a deep lungful of fresh air and went back to work.

The superliners left. But not without taking a souvenir. Francis Burbank went with them. His removal notice was on the first ship, and Don

Channing's appointment as Director of Interplanetary Communications was on the second.

Happily he entered the Director's office once more. He carried with him all the things that he had removed just a few short weeks before. This time he was coming to stay.

Arden entered the office behind him. "Home again?" she asked.

"Yop," he grinned at her. "Open file B, will you, and break out a container of my favorite beverage?"

"Sure thing," she said.

There came a shout of glee. "Break out four glasses," she was told from behind. It was Walt Franks and Joe.

It was Arden that proposed the toast. "Here's to a closed corporation," she said. They drank on that.

She went over beside Don and took his arm. "You see?" she said, looking up into his eyes. "We aren't the same. Things have changed since Burbank came, and went. Haven't they?"

"They have," laughed Channing. "And now that you are my secretary, it is no longer proper for you to shine up to me like that. People will talk."

"What's he raving about?" asked Joe.

Channing answered. "It is considered bad taste for a secretary to make passes at her boss. Think of his wife and kids."

"You have neither."

"Maybe so. But it is still not proper for a secretary to—"

"You can't call me a secretary in that tone of voice," snapped Arden. "I quit! I resign! I refuse to be secretary to a man like you!"

Channing looked helplessly at Franks.

Walt looked at Arden, saw what was in her eyes, and told Channing: "See if you can wriggle out of what comes next!" He took Joe by the arm and said: "Joe, now that the ban is off, may I buy you a drink?"

And Joe answered: "It is a beautiful night out, isn't it?"

It is always a beautiful night on Venus Equilateral. The stars shine forever in a sky that holds a molten ball. Sol flares endlessly in an absolutely black, star-studded sky. There is no moon. The air is always soft and warm and unchanging.

And at the moment that Channing was finding out why Arden resigned, a little man of Northern Venus handed a message to the operator in the International Hotel in Detroit, Michigan. It went out on the land wires to Hawaii; to Luna; to Venus Equilateral; to the rotating relay stations that circle Venus five hundred miles above the planet; down through the raging heaviside layer to Northern Landing; and across the Palanortis Country to Yoraleen.

Channing was still investigating his secretary's resignation when Korvus, the Magnificent, read:

TYPE 56-XXD FLIER F. O. B. DETROIT, MICHIGAN,
TODAY. WILL BE DELIVERED ON NEXT FREIGHT SHIP.

WILNEDA.

THE END.



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